

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-21** **Date: 20.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

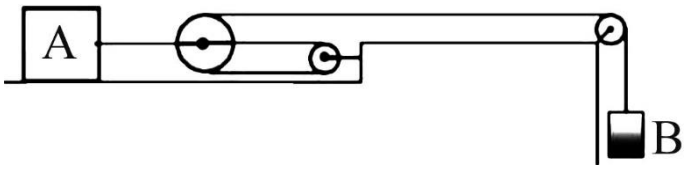
1. Match List I with List II

	List I		List II
a)	Pressure gradient	I)	$M^1L^2T^{-2}$
b)	Work function (ϕ)	II)	$M^1L^1T^{-3}A^{-1}$
c)	Moment of inertia	III)	$ML^{-2}T^{-2}$
d)	Electric field	IV)	ML^2T^0

A) $a-IV, b-II, c-III, d-I$ B) $a-III, b-I, c-IV, d-II$
C) $a-III, b-IV, c-II, d-I$ D) $a-III, b-I, c-II, d-IV$

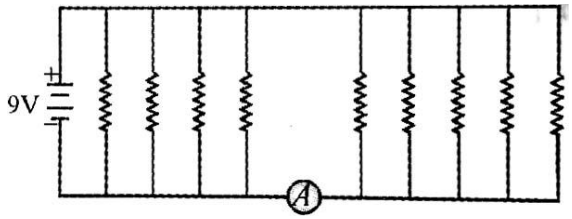
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2. A block A has a velocity of 0.6 m/s to the right, determine the velocity of cylinder B, at the same instant



A) 1.2 m/s B) 2.4 m/s C) 1.8 m/s D) 3.6 m/s

3. If each resistance in the figure is of 9Ω then reading of ammeter is



A) 5A B) 8A C) 2A D) 9A

4. The Young's modulus of a rubber string of 8cm long and density $1.5kg/m^3$ is $5\times10^8 N/m^2$, and is suspended from the ceiling in a room. The increase in length due to its own weight will be , take g = 10m/s^2

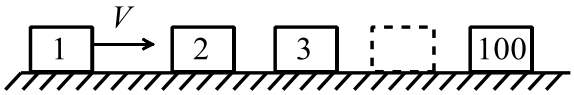
A) $9.5\times10^{-5}m$ B) $9.6\times10^{-11}m$ C) $9.6\times10^{-3}m$ D) 9.6 m

5. Photoelectric threshold wavelength of silver is $\lambda = 3800\,\mathring{A}$. Ultraviolet light of wavelength $\lambda = 2600\,\mathring{A}$ is incident on silver surface. What will be the value of work function?

A) $5.23\times10^{-19}J$ B) $6.5\times10^{-18}J$

C) $6.5\times10^{-19}J$ D) $5.5\times10^{-10}J$

6. There are hundred identical blocks equally spaced on a frictionless track as shown in the figure. Initially, all the blocks are separate. Each collision is perfectly inelastic. The final velocity will be

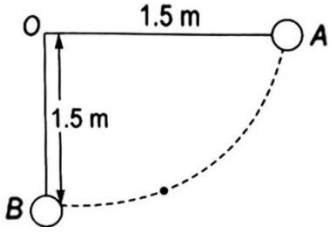


A) $\frac{V}{99}$ B) $\frac{V}{100}$ C) Zero D) V

7. The bob of a pendulum is released from a horizontal position A as shown in the figure. If the length of the pendulum is 1.5m, then the speed with which the bob arrives at the lower most point B. Given that it dissipated 5% of its initial gravitational potential energy with respect to position B, against air resistance?

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A) 6.0 m/s B) 6.5 m/s C) 4.5 m/s D) 5.3 m/s

8. If c is the speed of electromagnetic waves in vacuum, its speed in a medium of dielectric constant k and relative permeability μ is

A) $v=\frac{c}{\sqrt{\mu k}}$ B) $v=\frac{1}{\sqrt{\mu k}}$ C) $v=\frac{k}{\sqrt{\mu c}}$ D) $v=c\sqrt{\mu k}$

9. Two concentric coils, each of radius equal to $2\pi\ cm$, are placed at right angles to each other. Current of 3 and 4 ampere respectively, are flowing through the two coils. The magnetic induction in Wbm^{-2} at the centre of the coils will be $\left[\mu_0=4\pi\times10^{-7}Wb\left(Am\right)^{-1}\right]$

A) 5×10^{-5} B) 7×10^{-5} C) 12×10^{-5} D) 10^{-5}

10. First diffraction minimum due to a single slit diffraction is at $\theta=30°$ for a light of wavelength $6000\overset{o}{\text{A}}$. The width of the slit is

A) $1\times10^{-6}cm$ B) $1.2\times10^{-6}m$ C) $2\times10^{-6}cm$ D) $2.4\times10^{-6}m$

11. A transformer has 400 primary turns and 300 secondary turns. If the operating voltage for the load connected to the secondary is measured to be 300V, what is the voltage supplied to the primary?

A) 400 V B) 200 V C) 100 V D) 300 V

12. A particle of mass M is placed at the centre of a uniform spherical shell of equal mass 'M' and radius a. Find the gravitational potential at a point P at a distance $a/2$ from the centre

A) $-\frac{5GM}{a}$ B) $\frac{5GM}{a}$ C) $\frac{3GM}{a}$ D) $-\frac{3GM}{a}$

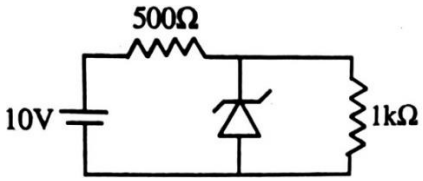
13. A silver wire has a resistance of 2.1Ω at $27.5^{o}C$ and a resistance of 2.7Ω at $100^{o}C$. Determine the temperature coefficient of resistivity of silver (nearly)

A) $0.049/°C$ B) $0.0049/°C$ C) $0.004/°C$ D) $0.039/°C$

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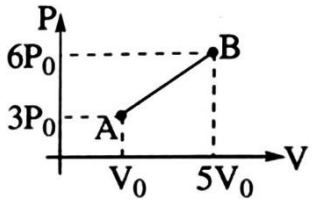
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14. The circuit of a simple constant voltage supply using a Zener diode. The constant voltage available across the Zener diodes is 5V. The current flowing through the $1k\Omega$ load is



A) 20 mA B) 15 mA C) 25 mA D) 5 mA

15. One mole of a monoatomic ideal gas undergoes the process $A \to B$ in the given P-V diagram. The specific heat for this process is



A) $\frac{3R}{2}$ B) $\frac{13R}{6}$ C) $\frac{5R}{2}$ D) 2R

16. Electric field on the axis of a small electric dipole at a distance r is $\vec{E}_1$ and $\vec{E}_2$ at a distance of 2r on a line of perpendicular bisector. Then

A) $\vec{E}_2 = -\vec{E}_1/8$ B) $\vec{E}_2 = -\vec{E}_1/16$ C) $\vec{E}_2 = -\vec{E}_1/4$ D) $\vec{E}_2 = -\vec{E}_1/8$

17. A uniform magnetic field exists in region given by $\vec{B} = 3\hat{i} + 4\hat{j} + 5\hat{k}$. A rod of length $5\,m$ placed along y-axis is moved along x-axis with constant speed $1m/\sec$. Then induced emf in the rod will be

A) 0 B) $25V$ C) $10V$ D) $15V$

18. If the velocity of a rotating body of mass 50kg is given by $\vec{v} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and radius vector $\vec{r} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, then its angular momentum is

A) $50\hat{i} + 50\hat{j} + 50\hat{k}$ B) $50\hat{i} + 100\hat{j} + 50\hat{k}$

C) $50\hat{i} + 50\hat{j} + 100\hat{k}$ D) $-50\hat{i} + 100\hat{j} - 50\hat{k}$

19. The temperature to which the hydrogen at 327°C be cooled at constant pressure, so that the root mean square velocity of its molecules becomes half of its previous value is

A) $-123°C$ B) $23°C$ C) $-100°C$ D) $0°C$

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20. An AC source of angular frequency ω is fed across a resistor R and a capacitor C in series. The current registered is I. If now the frequency of source is changed to $\omega/3$ (but maintaining the same voltage), the current in the circuit is found to be halved. The ratio of reactance to resistance at the original frequency ω is

A) $\sqrt{\frac{3}{5}}$ B) $\sqrt{\frac{5}{3}}$ C) $\frac{3}{5}$ D) $\frac{5}{3}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. The focal length of a thin biconvex lens is 20cm. When an object is moved from a distance of 25cm in front of it to 50cm, the magnification of its image changes from m_{25} to m_{50}. The ratio $\frac{m_{25}}{m_{50}}$ is _____

22. A ball is thrown from the top of a tower in vertically upward direction. Velocity at a point h meter below the point of projection is twice of the velocity at a point h meter above the point of projection. The maximum height reached by the ball above the top of tower is $\frac{nh}{3}$, then the value of n is _____

23. A parallel plate capacitor of capacitance C is connected to a battery and is charged to a potential difference V. Another capacitor of capacitance 2C is similarly charged to a potential difference 2V. The charging battery is now disconnected and the capacitors are connected in parallel to each other in such a way that the positive terminal of one is connected to the negative terminal of the other. The final energy of the configuration is $\frac{x}{y}CV^2$, then the value of $x+y=$ _____

24. For a certain organ pipe three successive resonance frequencies are observed at 425Hz, 595Hz and 765Hz respectively. If the speed of sound in air is 340m/s, then the length of the pipe is _____m

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25. A copper disc of diameter 20cm makes 1200rpm about its natural axis kept parallel to a uniform magnetic field of $10^{-2}T$. The potential difference between the centre and edge of the disc is $n\pi mV$, then the value of n is _____

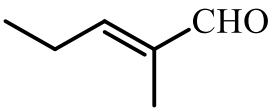
CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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26. Which of the following is crossed aldol condensation product

A) $PhCHO + PhCH_2OH$ B) $C_6H_5 - CH = CH - \overset{\overset{O}{\|}}{C} - C_6H_5$

C) $HCOO^{\ominus} + PhCH_2OH$ D) 

27. 20% of a first order reaction is completed in 20 min. Then 50% of that reaction is completed in ____ min (nearest value) (Given: $\log_{10} 2 = 0.3010$)

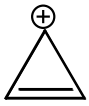
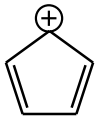
A) 50 min B) 72 min C) 56 min D) 62 min

28. $B \xleftarrow[(ii)H_2O]{(i)\ LiAlH_4} A \xrightarrow{Br_2+NaOH} C$

When the product 'B' and 'C' are homologues, then which of the following is true?

A) 'A' is an alcohol B) 'C' is a secondary amine
C) 'B' is a primary amine D) 'B' is an ester

29. Which of the following molecule/species is most unstable?

A)  B)  C)  D) Tropylium cation

30. Paramagnetic species are

A) Nd^{+3} and Eu^{+3} B) Yb^{+2} and Lu^{+3}
C) La^{+3} and Ce^{+4} D) Zn^{+2} and Sc^{+3}

31. S^{-2} ion can be identified by using sodium nitroprusside giving a violet colored complex compound. The geometry of the anion of the complex is

A) Trigonal bipyramidal B) Tetrahedral
C) Square planar D) Octahedral

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32. An amine X, reacts with $NaNO_2$ and HCl, at a temperature of 273-278K, and undergoes diazotization. 'X' can also undergo coupling reaction with the above diazo compound (BDS). Then 'X' is

A) Aliphatic primary amine
B) Aliphatic secondary amine
C) 2°-Aromatic amine
D) Aromatic primary amine

33. Ethyl benzene can be prepared from acetophenone by a reduction reaction. That reaction can be named as

A) Etard reaction
B) Friedel Craft's alkylation
C) Wolff-Kishner reaction
D) Cannizzaro reaction

34. Given below are half-cell reaction

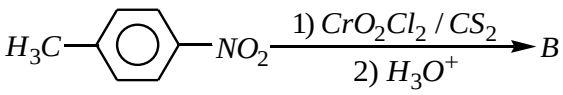
$A^{+2} + 2e^- \rightarrow A;\ E^0 = -0.56V$

$A^{+3} + e^- \rightarrow A^{+2};\ E^0 = +1.48V$

The E^0 for $3A^{+2} \rightarrow A + 2A^{+3}$ will be

A) $-2.04V$, reaction will not occur
B) $2.04\,V$, reaction will occur
C) $-0.92\,V$, reaction will not occur
D) $0.92\,V$, reaction will occur

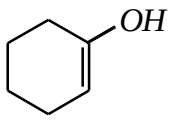
35. $(C_6H_5CH_2)_2Cd + CH_3COCl \longrightarrow A$

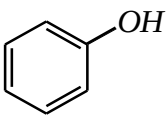


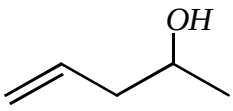
The true statement regarding the carbon compounds 'A' and 'B' is

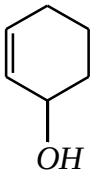
A) 'A' cannot give iodoform test
B) 'B' gives iodoform test
C) 'A' can give silver mirror test
D) 'B' is an aromatic aldehyde formed through a chromium complex

36. Which of the following contains allylic alcohol group?

A) 

B) 

C) 

D) 

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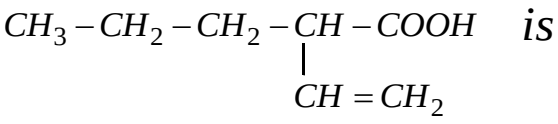
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37. Match List-I with List-II

	List I (Molecule/Ion)		List II (Shape)
a)	$PCl_{5(g)}$	I)	Linear
b)	D_2O	II)	Trigonal planar
c)	I_3^{-1}	III)	Trigonal bipyramidal
d)	NO_3^{-1}	IV)	Bent

A) a-I, b-II, c-III, d-IV
B) a-III, b-IV, c-I, d-II
C) a-III, b-IV, c-II, d-I
D) a-IV, b-III, c-I, d-II

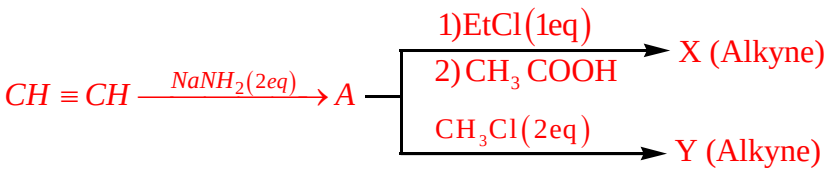
38. The correct IUPAC name of



is

A) 2-Ethenyl pentanoic acid
B) 3-Propyl-1-butenoic acid
C) 2-Propyl-3-butenoic acid
D) 2-Propyl butane-oic acid

39. Higher alkynes can be prepared from lower alkynes by using sodamide, followed by nucleophilic substitution reaction with alkyl halide



X & Y are
A) functional isomers
B) ring chain isomers
C) positional isomers
D) tautomers

40. Name and symbol, according to IUPAC nomenclature of an element which belongs to 7th period and 16th group is
A) Ununhexium, Lv
B) Ununbium, Cn
C) Unniltrium, Lr
D) Unnilhexium, Sg

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41. Given below are two statements

Statement-I : In hydrogen atom, the ratio for radius of first three orbits is 1 : 4 : 9

Statement-II : In hydrogen atom, the ratio for the energy gap between first three orbits is 1:3:5

A) Both Statement-I and Statement-II are correct

B) Both Statement-I and Statement-II are incorrect

C) Statement-I is correct, Statement-II is incorrect

D) Statement-I is incorrect, Statement-II is correct

42. The ratio for the number of Pi bonds in Toluene, Naphthalene, Biphenyl and Phenanthrene is

A) 3 : 5 : 7 : 9

B) 3 : 5 : 6 : 7

C) 3 : 5 : 6 : 6

D) 3 : 5 : 5 : 6

43. Given below are two statements

Statement-I : Brown ring test is a confirmatory test for NO_3^- ions

Statement-II : The brown ring is due to the formation of iron cation complex with $+3$ charge

A) Both Statement-I and Statement-II are correct

B) Both Statement-I and Statement-II are incorrect

C) Statement-I is correct, Statement-II is incorrect

D) Statement-I is incorrect, Statement-II is correct

44. The pair of compounds in which both the compounds give positive test with Tollen's reagent is

A) Glucose and Sucrose

B) Fructose and Sucrose

C) Acetophenone and Hexanal

D) Glucose and Fructose

45. The gas liberated, on warming a salt with $dil.H_2SO_4$, turns acidified $K_2Cr_2O_7$ solution into green. Then the salt is

A) Sulphate salt

B) Carbonate salt

C) Sulphite salt

D) Nitrate salt

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46. 500mL of 0.1M $NaOH$ solution is mixed with 250mL of 0.2M $NaOH$ along with 'x'g of pure $NaOH$ and the solution is made up to l lit. by adding water to make it 0.4M. The value of 'x' is _____

47. In Lassaigne's test, nitrogen is detected with the help of prussian blue colored complex. The number of iron atoms and total number of atoms in that one molecule are 'x' and 'y' respectively. Then $(x+y)$ is _____ (Assume all elements in the complex as atoms)

48. Oxidizing agent : Acidified $K_2Cr_2O_7$
Reducing agent : Sodium oxalate
In the balanced redox reaction with simplest coefficients, 'x' moles of dichromate reacts with three moles of oxalate 'z' moles of CO_2 is obtained, then the value of $(x+z)$ is

49. The difference between boiling point and melting point of 0.2 molal glucose aqueous solution is 'x', then the value of '10x' is
($K_f = 2K.kg\ mole^{-1}$ and $K_b = 0.5K.kg\ mol^{-1}$)

50. At $25°C$ K_{sp} of $Mg(OH)_2 = 4\times10^{-12}$. Then at $25°C$ P^H of saturated $Mg(OH)_2$ solution is 'k', then the value of '10k' is $(\log 2 = 0.3)$

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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51. A line L_1 passes through the point $(2,3)$ and makes 45° with positive x-axis. Another line L_2 passes through the point $(-1,0)$ and makes 15° with the line L_1. The equation of line L_2 is

A) $\sqrt{3}y = x-1$ B) $x+\sqrt{3}y+1=0$ C) $y=\sqrt{3}(x+1)$ D) $y+\sqrt{3}x+1=0$

52. Let S_n denotes sum of n terms of an A.P. If for one A.P, $S_n = 2n^2+n$ and for another A.P, $S_n = \frac{n(3n+1)}{2}$, then for the series of common terms of two A.P's, $S_n =$

A) $3n^2+2n$ B) $6n^2+5n$ C) $6n^2+6n$ D) $3n^2+8n$

53. If $27iz^3+18z^2-12z+8i=0$, then $|z| =$

A) 2 B) $\frac{3}{2}$ C) $\frac{2}{3}$ D) 4

54. If $\bar{a}=\hat{i}-\hat{j}+\hat{k}$, $\bar{b}=3\hat{i}-\hat{k}$ and $\bar{c}$ is a vector such that $(\bar{a}+\bar{b})\times\bar{c}=\bar{c}\times(-2\bar{a}+\bar{b})$ and $(3\bar{a}+\bar{b}).\bar{c}=42$, then $|\bar{c}| =$

A) $2\sqrt{15}$ B) $\sqrt{15}$ C) $2\sqrt{35}$ D) $\sqrt{35}$

55. The maximum volume of cone having slant height of length $3\sqrt{3}$ is

A) 72π B) 36π C) 18π D) 9π

56. $\lim\limits_{n\to\infty}\left\{\frac{1}{n^2}\sin\left(\frac{1+n^2}{n^2}\right)+\frac{2}{n^2}\sin\left(\frac{4+n^2}{n^2}\right)+\frac{3}{n^2}\sin\left(\frac{9+n^2}{n^2}\right)+.....+\frac{2}{n}\sin 5\right\} =$

A) $\cos 2\sin 3$ B) $\sin 2\sin 3$ C) $\sin 2\cos 3$ D) $\cos 2\cos 3$

57. If $f(x)=\sqrt{x+2\sqrt{2x-4}}+\sqrt{x-2\sqrt{2x-4}}$, then f is differentiable for $x\in$

A) $[2,\infty)$ B) $[4,\infty)$ C) $[2,\infty)-\{2\}$ D) $[2,\infty)-\{4\}$

58. Area bounded by the curve $y=\sqrt{x}$, $x=2y+3$, x-axis in first quadrant is

A) 9 B) $\frac{9}{2}$ C) 3 D) $\frac{3}{2}$

APEX JEE-MAIN 2025

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59. If $y=y(x)$ be the solution of differential equation $\cos x dy-\sin x dx=x dx-dy$ and $y\left(\frac{\pi}{2}\right)=\pi$, then $y\left(\frac{3\pi}{2}\right)=$

A) $\frac{\pi}{2}$ B) $-\frac{\pi}{2}$ C) π D) $-\pi$

60. If the image of the point $P(1,0,0)$ in the line $\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+10}{8}$ is (α,β,γ) then $\alpha+\beta+\gamma=$

A) 11 B) –11 C) 7 D) –7

61. A bag contains 4 tickets with numbers 0, 1, 10, 11. From the bag, five tickets are drawn one by one with replacement. The probability that the sum of the numbers on the tickets drawn being 23 is

A) $\frac{25}{256}$ B) $\frac{13}{128}$ C) $\frac{7}{64}$ D) $\frac{17}{128}$

62. The domain of the function $\cos^{-1}\left(\frac{x}{2}-3\right)+\sqrt{49-x^2}+\log\left(x^2-11x+30\right)$ is $[a,b)\cup(c,d]$, then $a+b+c+d=$

A) 15 B) 22 C) 21 D) 16

63. If the system of equations $3x-2y+z=b$, $5x-8y+9z=3$, $2x+y+az=-1$ has infinite number of solutions, then

A) $ab=1$ B) $a+b=0$ C) $a=9b$ D) $9b+a=0$

64. Consider two circles $x^2+y^2-2x-6y-6=0$, $x^2+y^2-8x+2y+8=0$. If n_1 is number of direct common tangents and n_2 is number of indirect (transverse) common tangents, then the equation of locus of the point equidistant from the points $(n_1,0)$ and $(0,n_2)$ is

A) $x=1$ B) $x=y$ C) $4x-2y-3=0$ D) $y=1$

65. For the ellipse, length of latus rectum is equal to distance from centre to one focus. If its eccentricity is of the form $\frac{a+\sqrt{b}}{c}$ (b is prime number), then $\frac{a+b}{c}=$

A) 2 B) 4 C) 6 D) 8

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66. Consider the following distribution

x	:	5	7	9	11	13	15	17
f	:	2	4	6	8	10	12	8

The mean deviation about median of the distribution is

A) 1.69 B) 2.72 C) 0.82 D) 3.1

67. If $\Delta=\begin{vmatrix} 2\cos^2 x & \sin 2x & -\sin x \\ \sin 2x & 2\sin^2 x & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix}$, then $\int_0^{\pi/2}\left(\Delta+\Delta^1\right)dx$ is $\left(\Delta^1=\frac{d}{dx}(\Delta)\right)$

A) $\frac{\pi}{4}$ B) $\frac{\pi}{2}$ C) 2π D) π

68. Δ_1 is the area of the parallelogram having adjacent sides $\bar{a}=3\hat{i}-\hat{k}$ and $\bar{b}=\hat{i}+2\hat{j}$ and Δ_2 is the area of the parallelogram having diagonals $\bar{p}=3\hat{i}+\hat{j}-2\hat{k}$ and $\bar{q}=\hat{i}-3\hat{j}+4\hat{k}$, then $\Delta_1^2+\Delta_2^2=$

A) 85 B) 100 C) 116 D) $\frac{1241}{4}$

69. The equation $2\tan^2 x-5\sec x-1=0$ has exactly 7 distinct values of x in $\left[0,\frac{n\pi}{2}\right], n\in N$. If n_1 and n_2 are the minimum and maximum values of 'n' respectively, then $n_1+n_2=$

A) 15 B) 26 C) 28 D) 30

70. The value of , $\lim_{x\to 0}\frac{\int_0^{x^2}\sec\theta^3 d\theta}{\tan^2 x}$ is

A) 1 B) –1 C) 2 D) 3

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. The shortest distance between lines $\frac{x+2}{1}=\frac{y}{-2}=\frac{z-5}{2}$ and $\frac{x-4}{1}=\frac{y-1}{2}=\frac{z+3}{0}$ is p. The perpendicular distance from the point $(0,3,-2)$ to the line $\frac{x+1}{2}=\frac{y-2}{1}=\frac{z-1}{-2}$ is q. The value of $p^2+q^2=$_____

72. If $\int_{-4}^{1}\|[2x+3]\|\,dx=\alpha$, then $4\alpha=$ ($[x]$ denotes greatest integer function) _____

73. Number of irrational terms in the expansion of $\left(2^{1/3}+5^{1/6}\right)^{1000}$ is _____

74. Let $y=y(x)$ be the solution of differential equation $x(x-1)\frac{dy}{dx}-(x-2)y=x^3(2x-1)$. If $y(2)=0$, then $y(3)=$ _____

75. The sequence of natural numbers is divided as (1, 2), (3, 4, 5, 6), (7, 8, 9, 10, 11, 12), …… The sum of the numbers in 10th bracket is _____

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